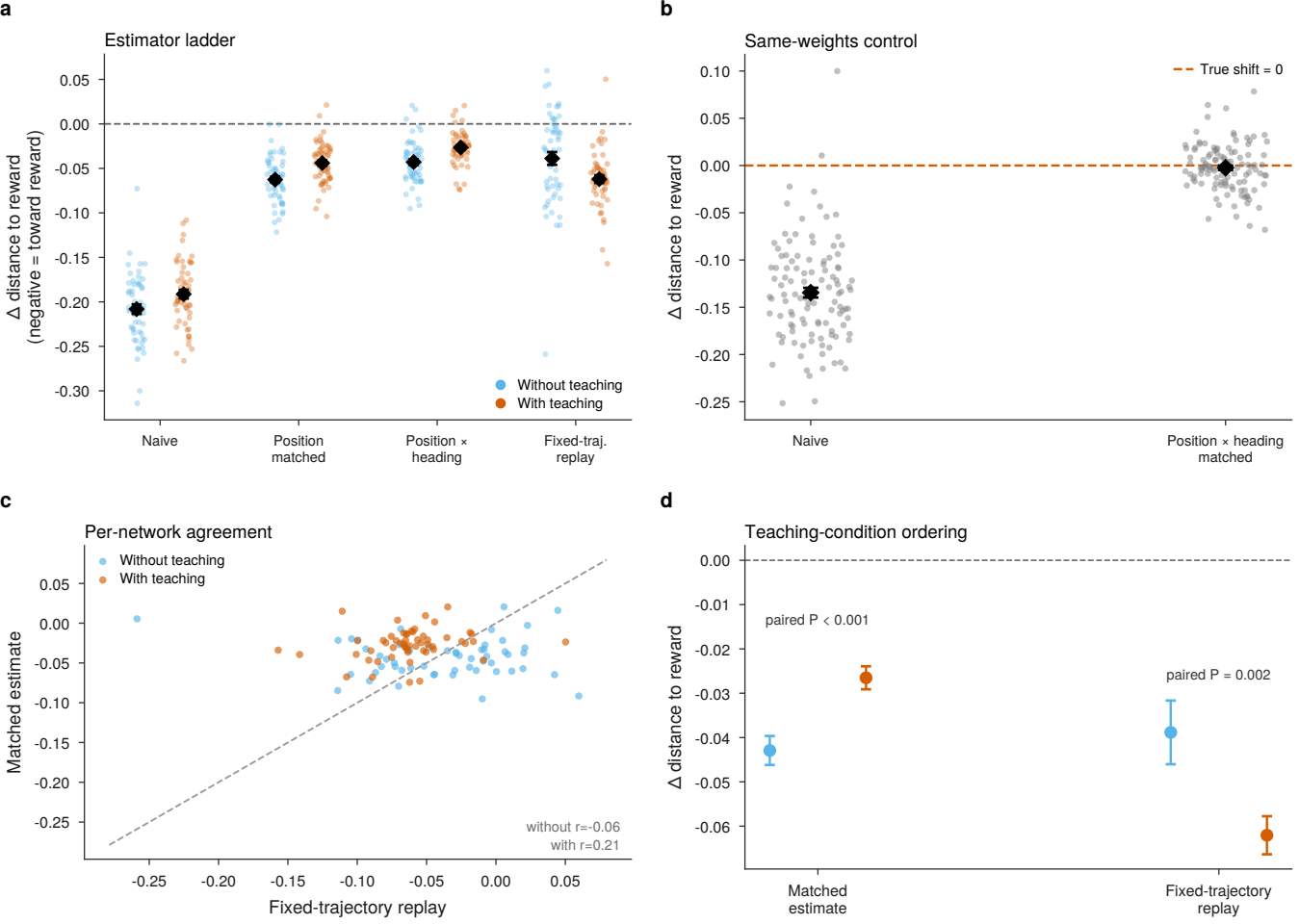

a
Estimator ladder
Δ distance to reward
(negative = toward reward)
0.05
0.00
-0.05
-0.10
-0.15
-0.20
-0.25
-0.30
Without teaching
With teaching
Naive
Position matched
Position × heading
Fixed-traj. replay
b
Same-weights control
True shift = 0
Δ distance to reward
0.10
0.05
0.00
-0.05
-0.10
-0.15
-0.20
-0.25
Naive
Position × heading matched
c
Per-network agreement
Without teaching
With teaching
Matched estimate
0.05
0.00
-0.05
-0.10
-0.15
-0.20
-0.25
without r=-0.06
with r=0.21
-0.25
-0.20
-0.15
-0.10
-0.05
0.00
0.05
Fixed-trajectory replay
d
Teaching-condition ordering
Δ distance to reward
0.00
-0.01
-0.02
-0.03
-0.04
-0.05
-0.06
paired P < 0.001
paired P = 0.002
Matched estimate
Fixed-trajectory replay